

Victoria Nguyen

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EDUCATION

University of Michigan - Ann Arbor

Bachelor of Science in Computer Science, Minor in Entrepreneurship

Ann Arbor, MI

May 2028

- **Relevant Coursework:** Data Structures and Algorithms, Discrete Math, Biomedical Innovation, Digital Product Design

TECHNICAL SKILLS

Languages: Python, C++, JavaScript (ES6+), TypeScript, SQL, HTML/CSS, Bash

Frameworks: React, FastAPI, Vite, Node.js, TailwindCSS, Express, React Native

Data & AI: Pandas, SDV (Synthetic Data Vault), FAISS (Vector Search), RAG, Plotly, D3.js, NumPy

Tools & Infrastructure: Git/GitHub, Docker, MongoDB, REST APIs, HL7 FHIR, Jupyter, Figma

WORK EXPERIENCE

Michigan Medicine - Department of Computational Medicine

Research Assistant - Genomic AI Development

Ann Arbor, MI

Jan 2026 – Present

- Developing a **Conversational AI Interface** for a genomic foundation model (*Nucleic Acids Res.* 2023), enabling researchers to query DNA binding predictions across the **3B+ base-pair human genome**.
- Building a **FastAPI** middleware layer to bridge **React** frontends with transformer model inference, reducing query latency for 50kb genomic windows by **40%** via optimized caching.
- Implementing an interactive **D3.js Genome Browser** that visualizes **transcription factor binding affinities** with real-time zooming, handling datasets of **10,000+ annotations** without rendering lag.
- Collaborating with PhD researchers to translate **state-of-the-art deep learning metrics** (AUC >0.95) into interpretability features, such as motif highlight maps, for non-technical biologists.

Michigan Medicine - Department of Pediatrics

Research Assistant - ABC Growing Kids Project

Ann Arbor, MI

2025 – Present

- Engineered automated data cleaning scripts in **Python (Pandas)** to process behavioral datasets, saving the research team **15+ manual hours per week** and eliminating transcription errors.
- Achieved **90%+ inter-rater reliability** on video behavioral coding using **Observer XT**, creating a high-fidelity "Ground Truth" dataset for training downstream predictive models.

Michigan Medicine University Hospital

Nurse Aide II

Ann Arbor, MI

2025 – 2026

- Leveraged direct clinical experience with **EPIC EMR** workflows to identify critical usability gaps, directly informing the **UI/UX architecture** of the Heartline Care Companion software project.
- Managed sensitive patient data entry for a 20-bed transplant unit, developing a strong understanding of **HIPAA compliance** and structured health data standards.

EMU Center for Health Disparities

Research Assistant - Asian American Health

Ann Arbor, MI

2024 – 2025

- Performed **quantitative analysis** on health survey data from **349 participants**, uncovering key disparities that drove strategic funding recommendations for local health initiatives.
- Designed **technical communication materials** to increase community engagement for **CMS Navigator programs**, translating complex federal health data into accessible public resources.

PROJECTS

SynthLab | React, FastAPI, Python, SDV, FAISS, RAG |

Dec 2025 – Present

- Architected a full-stack research platform using **React/Vite** (frontend) and **FastAPI** (backend), enabling secure synthetic data generation for clinical trials with a modern, responsive UI.
- Integrated **SDV (CTGAN, PAR)** models to synthesize longitudinal patient data that preserves **statistical correlations** and enforces physiological constraints (e.g., age, glucose levels).
- Developed a **Literature Intelligence** module using **FAISS vector embeddings** and LLM APIs to allow researchers to perform semantic search and automated summarization on local PDF libraries.
- Implemented **FHIR Interoperability** pipelines to export synthetic cohorts into **HL7 FHIR R4 JSON**, ensuring seamless integration with standard Electronic Health Records (EHR) systems.

Heartline Care Companion | React Native, Twilio, Python, Expo |

Oct 2025 – Nov 2025

- Built a mobile health app for transplant patients with **React Native**, implementing **Bluetooth LE** integration to sync vitals directly from medical devices to a structured JSON log.
- Integrated **Twilio SMS API** to trigger automated caregiver alerts when symptom algorithms detect a **20% deviation** from patient baselines, ensuring rapid clinical response.